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TỔNG KẾT PHASE B 24

HƯỚNG DẪN TRIỂN KHAI CHI TIẾT — PHASE B: BUSINESS LOGIC UPDATE

Dự án: SAP Bug Tracking Management System **Ngày:** 24/03/2026 | **Phiên bản:** 5.0 (Module Pool Integration) **Thời gian ước tính:** 2 ngày (25-27/03) **Yêu cầu:** Hoàn thành Phase A trước khi bắt đầu Phase B

> ⚠️ **QUAN TRỌNG:** Phải chạy report `Z_BUG_MIGRATE_STATUS` (Phase A, Bước A8) trước khi activate bất kỳ FM nào trong Phase B. Nếu không, status codes cũ (4=Fixed, 5=Closed) sẽ conflict với logic mới (4=Pending, 5=Fixed).

Bước B1: Mở rộng `Z_BUG_CHECK_PERMISSION`

Mục tiêu: Thêm Project permissions + user-project membership check.

Vào **SE37** → mở FM `Z_BUG_CHECK_PERMISSION` → **Change**.

Thêm IMPORTING parameter mới:

```
IV_PROJECT_ID TYPE ZDE_PROJECT_ID OPTIONAL
```

Source code cập nhật (thay thế toàn bộ):

```
FUNCTION z_bug_check_permission.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_USER) TYPE  ZDE_USERNAME
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID OPTIONAL
*"     VALUE(IV_ACTION) TYPE  CHAR20
```

```
*"     VALUE(IV_PROJECT_ID) TYPE  ZDE_PROJECT_ID OPTIONAL
*"  EXPORTING
*"     VALUE(EV_ALLOWED) TYPE  CHAR1
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_user    TYPE zbug_users,
        ls_bug     TYPE zbug_tracker,
        ls_project TYPE zbug_project,
        lv_count   TYPE i.

  " 1. Lấy thông tin User
  SELECT SINGLE * FROM zbug_users INTO @ls_user
    WHERE user_id = @iv_user AND is_del <> 'X'.

  IF sy-subrc <> 0.
    ev_allowed = 'N'.
    MESSAGE s029(zbug_msg) WITH iv_user INTO ev_message.
    RETURN.
  ENDIF.

  " 2. ĐẶC QUYỀN: Manager (M) có toàn quyền
  IF ls_user-role = 'M'.
    ev_allowed = 'Y'.
    RETURN.
  ENDIF.

  " 3. Lấy thông tin Bug (nếu có)
  IF iv_bug_id IS NOT INITIAL.
    SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
      WHERE bug_id = @iv_bug_id AND is_del <> 'X'.
  ENDIF.

  " 4. CHECK USER-PROJECT MEMBERSHIP (nếu có project context)
  DATA(lv_prj) = COND #( WHEN iv_project_id IS NOT INITIAL THEN iv_project_id
                         WHEN ls_bug-project_id IS NOT INITIAL THEN ls_bug-
project_id
                         ELSE '' ).
  IF lv_prj IS NOT INITIAL.
    SELECT COUNT(*) FROM zbug_user_project INTO @lv_count
      WHERE user_id = @iv_user AND project_id = @lv_prj.
    IF lv_count = 0.
      ev_allowed = 'N'.
      MESSAGE s004(zbug_msg) INTO ev_message.
      RETURN.
    ENDIF.
  ENDIF.

  " 5. RẼ NHÁNH KIỂM TRA QUYỀN THEO HÀNH ĐỘNG
```

```
  CASE iv_action.

    " === BUG ACTIONS ===
    WHEN 'CREATE'.
      IF ls_user-role = 'T'.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
        MESSAGE s005(zbug_msg) INTO ev_message.
      ENDIF.

    WHEN 'UPDATE_STATUS'.
      " Config Bug: Tester tự xử lý
      IF ls_bug-bug_type = 'F' AND ls_user-role = 'T' AND ls_bug-dev_id =
iv_user.
        ev_allowed = 'Y'.
      " Developer update bug do mình giữ
      ELSEIF ls_user-role = 'D' AND ls_bug-dev_id = iv_user.
        ev_allowed = 'Y'.
      " Tester update khi Bug mới tạo (Status 1)
      ELSEIF ls_user-role = 'T' AND ls_bug-status = '1'.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
        MESSAGE s006(zbug_msg) INTO ev_message.
      ENDIF.

    WHEN 'DELETE_BUG'.
      ev_allowed = 'N'.
      MESSAGE s006(zbug_msg) INTO ev_message.

    WHEN 'UPLOAD_REPORT'.
      IF ls_user-role = 'T' AND ls_bug-tester_id = iv_user.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
        MESSAGE s006(zbug_msg) INTO ev_message.
      ENDIF.

    WHEN 'UPLOAD_FIX'.
      IF ls_bug-bug_type = 'F' AND ls_user-role = 'T' AND ls_bug-dev_id =
iv_user.
        ev_allowed = 'Y'.
      ELSEIF ls_user-role = 'D' AND ls_bug-dev_id = iv_user.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
        MESSAGE s006(zbug_msg) INTO ev_message.
      ENDIF.
```

```
    WHEN 'UPLOAD_VERIFY'.
      IF ls_user-role = 'T'.
        ev_allowed = 'Y'.
      ELSE.
        ev_allowed = 'N'.
        MESSAGE s006(zbug_msg) INTO ev_message.
      ENDIF.

    " === PROJECT ACTIONS ===
    WHEN 'CREATE_PROJECT'.
      ev_allowed = 'N'.
      MESSAGE s007(zbug_msg) INTO ev_message.

    WHEN 'CHANGE_PROJECT'.
      ev_allowed = 'N'.
      MESSAGE s007(zbug_msg) INTO ev_message.

    WHEN 'DELETE_PROJECT'.
      ev_allowed = 'N'.
      MESSAGE s007(zbug_msg) INTO ev_message.

    WHEN 'VIEW_PROJECT'.
      ev_allowed = 'Y'.

    WHEN 'ADD_USER_PROJECT'.
      ev_allowed = 'N'.
      MESSAGE s007(zbug_msg) INTO ev_message.

    WHEN OTHERS.
      ev_allowed = 'N'.
      ev_message = 'Unknown action'.
  ENDCASE.

ENDFUNCTION.
```

Nhấn **Activate**.

✅ **Checkpoint: SE37** → `Z_BUG_CHECK_PERMISSION` → Test:

- `IV_USER` = (tester), `IV_ACTION` = `CREATE_PROJECT` → `EV_ALLOWED` = `N`
- `IV_USER` = (manager), `IV_ACTION` = `CREATE_PROJECT` → `EV_ALLOWED` = `Y`
- `IV_USER` = (tester), `IV_ACTION` = `CREATE`, `IV_PROJECT_ID` = `PRJ001` → check membership

Bước B2: Cập nhật `Z_BUG_CREATE`

Mục tiêu: Thêm `PROJECT_ID`, `SEVERITY`, user validation, project validation.

Vào **SE37** → mở FM `Z_BUG_CREATE` → **Change**.

Thêm IMPORTING parameters:

```
IV_PROJECT_ID TYPE ZDE_PROJECT_ID
IV_SEVERITY   TYPE ZDE_SEVERITY DEFAULT '4'
```

Source code cập nhật:

```
FUNCTION z_bug_create.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_TITLE) TYPE  ZDE_BUG_TITLE
*"     VALUE(IV_DESC) TYPE  ZDE_BUG_DESC
*"     VALUE(IV_MODULE) TYPE  ZDE_SAP_MODULE
*"     VALUE(IV_PRIORITY) TYPE  ZDE_PRIORITY DEFAULT 'M'
*"     VALUE(IV_DEV_ID) TYPE  ZDE_USERNAME OPTIONAL
*"     VALUE(IV_BUG_TYPE) TYPE  ZDE_BUG_TYPE OPTIONAL
*"     VALUE(IV_ATT_PATH) TYPE  ZDE_BUG_ATT_PATH OPTIONAL
*"     VALUE(IV_PROJECT_ID) TYPE  ZDE_PROJECT_ID
*"     VALUE(IV_SEVERITY) TYPE  ZDE_SEVERITY DEFAULT '4'
*"  EXPORTING
*"     VALUE(EV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(EV_SUCCESS) TYPE  CHAR1
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_bug     TYPE zbug_tracker,
        lv_number  TYPE numc10,
        lv_count   TYPE i.

  " ====== VALIDATION BLOCK ======

  " 1. Validate user exists + active
  SELECT COUNT(*) FROM zbug_users INTO @lv_count
    WHERE user_id = @sy-uname AND is_del <> 'X'.
  IF lv_count = 0.
    ev_success = 'N'.
    MESSAGE s029(zbug_msg) WITH sy-uname INTO ev_message.
    RETURN.
  ENDIF.

  " 2. Validate project exists + InProcess
  SELECT SINGLE project_status FROM zbug_project INTO @DATA(lv_prj_status)
    WHERE project_id = @iv_project_id AND is_del <> 'X'.
  IF sy-subrc <> 0.
```

```
  ev_success = 'N'.
  MESSAGE s021(zbug_msg) WITH iv_project_id INTO ev_message.
  RETURN.
ENDIF.
IF lv_prj_status <> '2'. " Must be InProcess
  ev_success = 'N'.
  MESSAGE s022(zbug_msg) INTO ev_message.
  RETURN.
ENDIF.

" 3. Validate user belongs to project
SELECT COUNT(*) FROM zbug_user_project INTO @lv_count
  WHERE user_id = @sy-uname AND project_id = @iv_project_id.
IF lv_count = 0.
  ev_success = 'N'.
  MESSAGE s004(zbug_msg) INTO ev_message.
  RETURN.
ENDIF.

" 4. Severity + Bug Type cross validation
IF iv_bug_type IS INITIAL.
  ls_bug-bug_type = 'C'. " Default Code bug
ELSE.
  ls_bug-bug_type = iv_bug_type.
ENDIF.

" Dump/VeryHigh/High severity Code bugs → force High priority
IF ( iv_severity = '1' OR iv_severity = '2' OR iv_severity = '3' )
   AND ls_bug-bug_type = 'C'.
  ls_bug-priority = 'H'.
ELSE.
  ls_bug-priority = iv_priority.
ENDIF.

" ====== BUG ID GENERATION ======

CALL FUNCTION 'NUMBER_GET_NEXT'
  EXPORTING
    nr_range_nr             = '01'
    object                  = 'ZNRO_BUG'
  IMPORTING
    number                  = lv_number
  EXCEPTIONS
    interval_not_found      = 1
    number_range_not_intern = 2
    object_not_found        = 3
    quantity_is_0           = 4
    quantity_is_not_1       = 5
    interval_overflow       = 6
```

```
    buffer_overflow         = 7
    OTHERS                  = 8.

IF sy-subrc <> 0.
  ev_success = 'N'.
  MESSAGE s011(zbug_msg) INTO ev_message.
  RETURN.
ENDIF.

CONCATENATE 'BUG' lv_number+3(7) INTO ev_bug_id.

" ====== FILL BUG RECORD ======

ls_bug-bug_id       = ev_bug_id.
ls_bug-title        = iv_title.
ls_bug-desc_text    = iv_desc.
ls_bug-sap_module   = iv_module.
ls_bug-tester_id    = sy-uname.
ls_bug-created_at   = sy-datum.
ls_bug-created_time = sy-uzeit.
ls_bug-ernam        = sy-uname.
ls_bug-erdat        = sy-datum.
ls_bug-erzet        = sy-uzeit.
ls_bug-project_id   = iv_project_id.
ls_bug-severity     = iv_severity.

" Evidence
IF iv_att_path IS NOT INITIAL.
  ls_bug-att_report = iv_att_path.
ENDIF.

" ====== WORKFLOW BRANCHING ======

IF ls_bug-bug_type = 'F'.
  " Config Bug: Tester tự xử lý
  ls_bug-status = '2'.
  ls_bug-dev_id = sy-uname.
ELSE.
  " Code Bug: New, chờ Auto-Assign
  ls_bug-status = '1'.
  ls_bug-dev_id = ''.
ENDIF.

" ====== DATABASE INSERT ======
INSERT zbug_tracker FROM ls_bug.

IF sy-subrc = 0.
  ev_success = 'Y'.
  MESSAGE s001(zbug_msg) WITH ev_bug_id INTO ev_message.
```

```
    " Log history
    CALL FUNCTION 'Z_BUG_LOG_HISTORY'
      EXPORTING
        iv_bug_id      = ev_bug_id
        iv_action_type = 'CR'
        iv_old_value   = ''
        iv_new_value   = iv_title
        iv_reason      = 'New bug created'.

    " Send email notification
    CALL FUNCTION 'Z_BUG_SEND_EMAIL'
      EXPORTING
        iv_bug_id = ev_bug_id
        iv_event  = 'CREATE'.

    COMMIT WORK AND WAIT.
  ELSE.
    ev_success = 'N'.
    MESSAGE s010(zbug_msg) INTO ev_message.
    ROLLBACK WORK.
  ENDIF.

ENDFUNCTION.
```

Nhấn **Activate**.

> ✅ **Checkpoint: SE37** → `Z_BUG_CREATE` → Test:

- Truyền `IV_PROJECT_ID` không tồn tại → "Project does not exist"
- Truyền phần tử Valid → Bug tạo thành công với `PROJECT_ID` + `SEVERITY`

Bước B3: Chuẩn hóa Status States — `Z_BUG_UPDATE_STATUS` (Full Rewrite)

Mục tiêu: Refactor `Z_BUG_UPDATE_STATUS` cho 9 status states mới với full validation.

Vào **SE37** → `Z_BUG_UPDATE_STATUS` → **Change**.

Bảng Transition hợp lệ:

From	→ Valid To
1	2 (assign), W (no dev)
W	2 (manager assign)
2	3 (dev start), R (reject)
3	5 (fixed), 4 (pending)
4	3 (resume)
5	6 (verify pass), 3 (fail)
6	7 (close)
7	*(terminal — no transition)*
R	2 (reassign)

Source code cập nhật (thay thế toàn bộ):

```
FUNCTION z_bug_update_status.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_NEW_STATUS) TYPE  CHAR1
*"     VALUE(IV_DEV_ID) TYPE  ZDE_USERNAME OPTIONAL
*"     VALUE(IV_REASON) TYPE  STRING OPTIONAL
*"  EXPORTING
*"     VALUE(EV_SUCCESS) TYPE  CHAR1
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_bug   TYPE zbug_tracker,
        lv_valid TYPE abap_bool VALUE abap_false.

  " ====== 1. READ CURRENT BUG ======
  SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
    WHERE bug_id = @iv_bug_id AND is_del <> 'X'.

  IF sy-subrc <> 0.
    ev_success = 'N'.
    MESSAGE s029(zbug_msg) WITH iv_bug_id INTO ev_message.
    RETURN.
  ENDIF.

  " ====== 2. CHECK BUG NOT CLOSED ======
  IF ls_bug-status = '7'.
    ev_success = 'N'.
    MESSAGE s033(zbug_msg) INTO ev_message.
    RETURN.
```

```
ENDIF.

" ====== 3. PERMISSION CHECK ======
CALL FUNCTION 'Z_BUG_CHECK_PERMISSION'
  EXPORTING
    iv_user   = sy-uname
    iv_bug_id = iv_bug_id
    iv_action = 'UPDATE_STATUS'
  IMPORTING
    ev_allowed = DATA(lv_allowed)
    ev_message = ev_message.

IF lv_allowed <> 'Y'.
  ev_success = 'N'.
  RETURN.
ENDIF.

" ====== 4. TRANSITION VALIDATION ======
CASE ls_bug-status.
  WHEN '1'.  " New
    IF iv_new_status = '2' OR iv_new_status = 'W'.
      lv_valid = abap_true.
    ENDIF.
  WHEN 'W'.  " Waiting
    IF iv_new_status = '2'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '2'.  " Assigned
    IF iv_new_status = '3' OR iv_new_status = 'R'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '3'.  " In Progress
    IF iv_new_status = '5' OR iv_new_status = '4'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '4'.  " Pending
    IF iv_new_status = '3'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '5'.  " Fixed
    IF iv_new_status = '6' OR iv_new_status = '3'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '6'.  " Resolved
    IF iv_new_status = '7'.
      lv_valid = abap_true.
    ENDIF.
  WHEN '7'.  " Closed — terminal
    lv_valid = abap_false.
```

```
  WHEN 'R'.  " Rejected
    IF iv_new_status = '2'.
      lv_valid = abap_true.
    ENDIF.
ENDCASE.

IF lv_valid = abap_false.
  ev_success = 'N'.
  DATA: lv_old_text TYPE char20, lv_new_text TYPE char20.
  lv_old_text = SWITCH #( ls_bug-status
    WHEN '1' THEN 'New'        WHEN 'W' THEN 'Waiting'
    WHEN '2' THEN 'Assigned'   WHEN '3' THEN 'InProgress'
    WHEN '4' THEN 'Pending'    WHEN '5' THEN 'Fixed'
    WHEN '6' THEN 'Resolved'   WHEN '7' THEN 'Closed'
    WHEN 'R' THEN 'Rejected' ).
  lv_new_text = SWITCH #( iv_new_status
    WHEN '1' THEN 'New'        WHEN 'W' THEN 'Waiting'
    WHEN '2' THEN 'Assigned'   WHEN '3' THEN 'InProgress'
    WHEN '4' THEN 'Pending'    WHEN '5' THEN 'Fixed'
    WHEN '6' THEN 'Resolved'   WHEN '7' THEN 'Closed'
    WHEN 'R' THEN 'Rejected' ).
  MESSAGE s018(zbug_msg) WITH lv_old_text lv_new_text INTO ev_message.
  RETURN.
ENDIF.

" ====== 5. EVIDENCE VALIDATION ======
" Chuyển sang Fixed (5): phải có TESTCASE file
IF iv_new_status = '5'.
  IF ls_bug-att_fix IS INITIAL.
    " Check GOS
    DATA: lt_files TYPE sbdst_files.
    CALL FUNCTION 'Z_BUG_GOS_LIST'
      EXPORTING iv_bug_id = iv_bug_id
      IMPORTING et_files  = lt_files.
    " Tìm file TESTCASE trong GOS
    DATA(lv_has_testcase) = abap_false.
    LOOP AT lt_files TRANSPORTING NO FIELDS
      WHERE comp_id CS 'TESTCASE' OR comp_id CS 'testcase'.
      lv_has_testcase = abap_true. EXIT.
    ENDLOOP.
    IF lv_has_testcase = abap_false.
      ev_success = 'N'.
      MESSAGE s016(zbug_msg) INTO ev_message.
      RETURN.
    ENDIF.
  ENDIF.
ENDIF.

" Chuyển sang Resolved (6): phải có CONFIRM file
```

```
IF iv_new_status = '6'.
  IF ls_bug-att_verify IS INITIAL.
    CALL FUNCTION 'Z_BUG_GOS_LIST'
      EXPORTING iv_bug_id = iv_bug_id
      IMPORTING et_files  = lt_files.
    DATA(lv_has_confirm) = abap_false.
    LOOP AT lt_files TRANSPORTING NO FIELDS
      WHERE comp_id CS 'CONFIRM' OR comp_id CS 'confirm'.
      lv_has_confirm = abap_true. EXIT.
    ENDLOOP.
    IF lv_has_confirm = abap_false.
      ev_success = 'N'.
      MESSAGE s017(zbug_msg) INTO ev_message.
      RETURN.
    ENDIF.
  ENDIF.
ENDIF.

" ====== 6. SPECIAL FIELD UPDATES ======
DATA(lv_old_status) = ls_bug-status.
ls_bug-status = iv_new_status.

" →2 (Assigned): Set DEV_ID
IF iv_new_status = '2' AND iv_dev_id IS NOT INITIAL.
  ls_bug-dev_id = iv_dev_id.
ENDIF.

" →6 (Resolved): Tester đã verify
IF iv_new_status = '6'.
  ls_bug-verify_tester_id = sy-uname.
ENDIF.

" →7 (Closed): Manager close
IF iv_new_status = '7'.
  ls_bug-closed_at   = sy-datum.
  ls_bug-approved_by  = sy-uname.
  ls_bug-approved_at  = sy-datum.
ENDIF.

" ====== 7. AUDIT FIELDS ======
ls_bug-aenam = sy-uname.
ls_bug-aedat = sy-datum.
ls_bug-aezet = sy-uzeit.

" ====== 8. DATABASE UPDATE ======
UPDATE zbug_tracker FROM ls_bug.

IF sy-subrc = 0.
  ev_success = 'Y'.
```

```
    MESSAGE s002(zbug_msg) WITH iv_bug_id INTO ev_message.

    " Log history
    CALL FUNCTION 'Z_BUG_LOG_HISTORY'
      EXPORTING
        iv_bug_id      = iv_bug_id
        iv_action_type = 'ST'
        iv_old_value   = lv_old_status
        iv_new_value   = iv_new_status
        iv_reason      = iv_reason.

    " Send email
    CALL FUNCTION 'Z_BUG_SEND_EMAIL'
      EXPORTING
        iv_bug_id = iv_bug_id
        iv_event  = 'STATUS_CHANGE'.

    COMMIT WORK AND WAIT.
  ELSE.
    ev_success = 'N'.
    MESSAGE s010(zbug_msg) INTO ev_message.
    ROLLBACK WORK.
  ENDIF.

ENDFUNCTION.
```

Nhấn **Activate**.

✅ **Checkpoint: SE37** → `Z_BUG_UPDATE_STATUS` → Test:

- Status `1` → `5` (Fixed) → FAIL "Invalid transition from New to Fixed"
- Status `1` → `2` (Assigned) → PASS
- Status `3` → `5` (Fixed) without TESTCASE file → FAIL "Upload TESTCASE file…"
- Status `6` → `7` (Closed) → PASS, CLOSED_AT + APPROVED_BY filled

---

Bước B4: GOS File Storage Integration

Mục tiêu: Tạo cơ chế upload file qua GOS thay local path.

Dùng BDS API:

Tạo FM mới `Z_BUG_GOS_UPLOAD` trong Function Group `ZBUG_FG`:

```
FUNCTION z_bug_gos_upload.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
```

```
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_FILE_PATH) TYPE  STRING
*"     VALUE(IV_FILE_TYPE) TYPE  CHAR20
*"  EXPORTING
*"     VALUE(EV_SUCCESS) TYPE  CHAR1
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: lt_file_content TYPE TABLE OF soli,
        lv_file_size    TYPE i,
        lv_file_name    TYPE string.

  " 1. Đọc file từ frontend
  CALL FUNCTION 'GUI_UPLOAD'
    EXPORTING
      filename   = iv_file_path
      filetype   = 'BIN'
    IMPORTING
      filelength = lv_file_size
    TABLES
      data_tab   = lt_file_content
    EXCEPTIONS
      OTHERS     = 1.

  IF sy-subrc <> 0.
    ev_success = 'N'.
    MESSAGE s000(zbug_msg) WITH 'Failed to read' 'file from local' INTO
ev_message.
    RETURN.
  ENDIF.

  " 2. Lấy tên file (basename)
  DATA: lt_parts TYPE TABLE OF string.
  SPLIT iv_file_path AT '\' INTO TABLE lt_parts.
  READ TABLE lt_parts INTO lv_file_name INDEX lines( lt_parts ).

  " 3. Lưu vào BDS
  DATA: lo_bds       TYPE REF TO cl_bds_document_set,
        lt_files     TYPE sbdst_files,
        ls_file      TYPE sbdst_file,
        lt_signature TYPE sbdst_signature,
        ls_signature TYPE sbdst_signat.

  CREATE OBJECT lo_bds.

  ls_signature-prop_name  = 'DESCRIPTION'.
  ls_signature-prop_value = |{ iv_file_type }: { lv_file_name }|.
  APPEND ls_signature TO lt_signature.
```

```
  ls_file-doc_count = 1.
  ls_file-comp_count = 1.
  ls_file-comp_id = lv_file_name.
  ls_file-mimetype = 'application/octet-stream'.
  ls_file-comp_size = lv_file_size.
  APPEND ls_file TO lt_files.

  TRY.
    lo_bds->create_with_table(
      EXPORTING classname  = 'ZBUG_EVIDENCE'
                classtype  = 'OT'
                object_key = CONV #( iv_bug_id )
      CHANGING  components = lt_files
                content    = lt_file_content
                signature  = lt_signature ).
    COMMIT WORK.
    ev_success = 'Y'.
    ev_message = |File { lv_file_name } uploaded successfully|.
  CATCH cx_bds_kernel INTO DATA(lx_error).
    ev_success = 'N'.
    ev_message = lx_error->get_text( ).
    ROLLBACK WORK.
  ENDTRY.

ENDFUNCTION.
```

Tạo FM Z_BUG_GOS_LIST:

```
FUNCTION z_bug_gos_list.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"  EXPORTING
*"     VALUE(ET_FILES) TYPE  SBDST_FILES
*"----------------------------------------------------------------------

  DATA: lo_bds       TYPE REF TO cl_bds_document_set,
        lt_signature TYPE sbdst_signature.

  CREATE OBJECT lo_bds.

  TRY.
    lo_bds->get_info(
      EXPORTING classname  = 'ZBUG_EVIDENCE'
                classtype  = 'OT'
                object_key = CONV #( iv_bug_id )
      CHANGING  components = et_files
                signature  = lt_signature ).
  CATCH cx_bds_kernel.
```

```
    CLEAR et_files.
  ENDTRY.

ENDFUNCTION.
```

Nhấn **Activate** cả 2 FMs.

Bước B5: SmartForm Email Template

Mục tiêu: Tạo SmartForm `ZBUG_EMAIL_FORM` cho email notification.

Vào T-code **SMARTFORMS** → nhập `ZBUG_EMAIL_FORM` → **Create**.

Step 1: Form Interface

- IMPORTING:
  - `IV_BUG_ID TYPE ZDE_BUG_ID`
  - `IV_EVENT  TYPE CHAR20`
  - `IS_BUG    TYPE ZBUG_TRACKER`
  - `IV_EVENT_TEXT TYPE STRING`

Step 2: Global Definitions

- Global Data:
  - `GV_TITLE TYPE STRING`
  - `GV_BODY  TYPE STRING`

Step 3: Form Layout

- Pages and Windows → MAIN Window → Main Window
- Text Element **BUG_INFO:**

```
Bug ID:    &IS_BUG-BUG_ID&
Title:     &IS_BUG-TITLE&
Project:   &IS_BUG-PROJECT_ID&
Status:    &IS_BUG-STATUS&
```

Nhấn **Save** → **Activate**.

Bước B6: Cập nhật `Z_BUG_SEND_EMAIL`

Mục tiêu: Sửa FM email để dùng SmartForm + event-based recipients.

```
FUNCTION z_bug_send_email.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_EVENT) TYPE  CHAR20
*"----------------------------------------------------------------------
```

```
  DATA: ls_bug       TYPE zbug_tracker,
        lt_recip     TYPE TABLE OF zbug_users,
        lo_send_req  TYPE REF TO cl_bcs,
        lo_doc       TYPE REF TO cl_document_bcs,
        lo_sender    TYPE REF TO cl_sapuser_bcs,
        lo_recip     TYPE REF TO if_recipient_bcs,
        lv_subject   TYPE so_obj_des,
        lt_body      TYPE bcsy_text,
        ls_body      TYPE soli,
        lv_sent      TYPE os_boolean.

  " 1. Lấy thông tin bug
  SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
    WHERE bug_id = @iv_bug_id AND is_del <> 'X'.
  IF sy-subrc <> 0. RETURN. ENDIF.

  " 2. Xác định recipients theo event
  CASE iv_event.
    WHEN 'CREATE'.
      SELECT * FROM zbug_users INTO TABLE @lt_recip
        WHERE role IN ('M','D') AND is_del <> 'X'.
    WHEN 'ASSIGN' OR 'REASSIGN'.
      SELECT * FROM zbug_users APPENDING TABLE @lt_recip
        WHERE user_id = @ls_bug-dev_id AND is_del <> 'X'.
    WHEN 'STATUS_CHANGE'.
      SELECT * FROM zbug_users INTO TABLE @lt_recip
        WHERE ( user_id = @ls_bug-tester_id OR user_id = @ls_bug-dev_id )
          AND is_del <> 'X'.
    WHEN 'REJECT'.
      SELECT * FROM zbug_users INTO TABLE @lt_recip
        WHERE role = 'M' AND is_del <> 'X'.
  ENDCASE.

  IF lt_recip IS INITIAL. RETURN. ENDIF.

  " 3. Tạo email body (HTML)
  lv_subject = |[BugTracker] { iv_event }: { ls_bug-bug_id } - { ls_bug-title }|.

  ls_body-line = |<html><body>|.                            APPEND ls_body TO
lt_body.
  ls_body-line = |<h2>Bug Tracking Notification</h2>|.     APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Event:</b> { iv_event }</p>|.      APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Bug ID:</b> { ls_bug-bug_id }</p>|. APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Title:</b> { ls_bug-title }</p>|.  APPEND ls_body TO
lt_body.
```

```
  ls_body-line = |<p><b>Project:</b> { ls_bug-project_id }</p>|. APPEND ls_body
TO lt_body.
  ls_body-line = |<p><b>Status:</b> { ls_bug-status }</p>|. APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Developer:</b> { ls_bug-dev_id }</p>|. APPEND ls_body TO
lt_body.
  ls_body-line = |<p><b>Tester:</b> { ls_bug-tester_id }</p>|. APPEND ls_body TO
lt_body.
  ls_body-line = |</body></html>|.                          APPEND ls_body TO
lt_body.

  " 4. Gửi email qua CL_BCS
  TRY.
    lo_send_req = cl_bcs=>create_persistent( ).
    lo_doc = cl_document_bcs=>create_document(
      i_type    = 'HTM'
      i_text    = lt_body
      i_subject = lv_subject ).
    lo_send_req->set_document( lo_doc ).

    lo_sender = cl_sapuser_bcs=>create( sy-uname ).
    lo_send_req->set_sender( lo_sender ).

    LOOP AT lt_recip ASSIGNING FIELD-SYMBOL(<user>).
      IF <user>-email IS NOT INITIAL.
        lo_recip = cl_cam_address_bcs=>create_internet_address( <user>-email ).
        lo_send_req->add_recipient( lo_recip ).
      ENDIF.
    ENDLOOP.

    lo_send_req->set_send_immediately( 'X' ).
    lv_sent = lo_send_req->send( ).
    COMMIT WORK.
  CATCH cx_bcs INTO DATA(lx_bcs).
    " Log error but don't block main flow
    DATA(lv_err) = lx_bcs->get_text( ).
  ENDTRY.

ENDFUNCTION.
```

Bước B7: Cập nhật Soft Delete Logic

Mục tiêu: Đảm bảo hệ thống không xóa thực sự dữ liệu (Hard Delete) mà chỉ đánh dấu xóa (Soft Delete) qua field `IS_DEL = 'X'`.

Thao tác thực hiện: Bạn cần vào **SE37**, mở lần lượt 2 FM sau đây để sửa code:

1. FM `Z_BUG_GET` (Lấy thông tin Bug)

- **Mục tiêu:** Thêm điều kiện lọc `is_del <> 'X'`.

- **Source code:**

```
SELECT SINGLE * FROM zbug_tracker INTO @es_bug
  WHERE bug_id = @iv_bug_id
    AND is_del <> 'X'.

IF sy-subrc = 0.
  ev_success = 'Y'.
ELSE.
  ev_success = 'N'.
  ev_message = 'Bug not found or was deleted'.
ENDIF.
```

2. FM `Z_BUG_DELETE` (Đánh dấu xóa)

- **Mục tiêu:** Thay vì `DELETE`, ta dùng `UPDATE` field `IS_DEL`.
- **Source code:**

```
DATA: ls_bug TYPE zbug_tracker.

" 1. Kiểm tra bug tồn tại
SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
  WHERE bug_id = @iv_bug_id
    AND is_del <> 'X'.

IF sy-subrc <> 0.
  ev_success = 'N'.
  ev_message = 'Bug not found or already deleted'.
  RETURN.
ENDIF.

" 2. Thực hiện Soft Delete (Dùng @ và dấu phẩy cho Strict Mode)
UPDATE zbug_tracker
  SET is_del = 'X',
      aenam  = @sy-uname,
      aedat  = @sy-datum,
      aezet  = @sy-uzeit
  WHERE bug_id = @iv_bug_id.

IF sy-subrc = 0.
  COMMIT WORK.
  ev_success = 'Y'.
  CONCATENATE 'Bug' iv_bug_id 'marked as deleted' INTO ev_message SEPARATED BY
SPACE.
ELSE.
  ROLLBACK WORK.
  ev_success = 'N'.
  ev_message = 'Failed to delete bug'.
ENDIF.
```

Nhấn **Activate** cho cả 2 FM.

> ✅ **Checkpoint:** Bất kỳ query nào `SELECT FROM zbug_tracker/zbug_users` đều có `WHERE is_del <> 'X'`.

Bước B8: Cập nhật `Z_BUG_AUTO_ASSIGN`

Mục tiêu: Update FM auto-assign để filter IS_DEL + chỉ assign dev trong cùng project.

> ⚠️ FM `Z_BUG_AUTO_ASSIGN` đã tồn tại từ MVP nhưng cần update đáng kể.

Vào **SE37** → mở FM `Z_BUG_AUTO_ASSIGN` → **Change**.

Source code cập nhật (thay thế toàn bộ):

```
FUNCTION z_bug_auto_assign.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_MODULE) TYPE  ZDE_SAP_MODULE
*"  EXPORTING
*"     VALUE(EV_DEV_ID) TYPE  ZDE_USERNAME
*"     VALUE(EV_STATUS) TYPE  ZDE_BUG_STATUS
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_bug      TYPE zbug_tracker,
        lt_devs     TYPE TABLE OF zbug_users,
        ls_dev      TYPE zbug_users,
        lv_count    TYPE i,
        lv_min      TYPE i VALUE 99999,
        lv_best_dev TYPE zde_username,
        lv_uname    TYPE sy-uname.

  lv_uname = sy-uname.

  " 1. Lấy thông tin bug
  SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
    WHERE bug_id = @iv_bug_id AND is_del <> 'X'.
  IF sy-subrc <> 0.
    ev_status = 'N'.
    ev_message = 'Bug not found'.
    RETURN.
  ENDIF.
```

```
" 2. Chỉ auto-assign cho bug ở trạng thái NEW (1)
IF ls_bug-status <> '1'.
  ev_status = 'N'.
  ev_message = 'Bug is not in NEW status'.
  RETURN.
ENDIF.

" 3. Tìm các Dev trong Project (ZBUG_USER_PROJEC)
SELECT u~user_id FROM zbug_users AS u
  INNER JOIN zbug_user_projec AS up ON u~user_id = up~user_id
  INTO TABLE @lt_devs
  WHERE u~role = 'D'
    AND u~is_del <> 'X'
    AND up~project_id = @ls_bug-project_id.

IF lt_devs IS INITIAL.
  " Không có Dev -> Chuyển sang trạng thái Waiting (W)
  UPDATE zbug_tracker SET status = 'W', aenam = @lv_uname
    WHERE bug_id = @iv_bug_id.
  COMMIT WORK.
  ev_status = 'W'.
  ev_message = 'No dev available - status set to Waiting'.
  RETURN.
ENDIF.

" 4. Tìm Dev ít việc nhất (Round-robin)
LOOP AT lt_devs INTO ls_dev.
  SELECT COUNT(*) FROM zbug_tracker INTO @lv_count
    WHERE dev_id = @ls_dev-user_id
      AND is_del <> 'X'
      AND status IN ('2','3').
  IF lv_count < lv_min.
    lv_min = lv_count.
    lv_best_dev = ls_dev-user_id.
  ENDIF.
ENDLOOP.

" 5. Cập nhật Assign
UPDATE zbug_tracker SET dev_id = @lv_best_dev,
                        status = '2',
                        aenam  = @lv_uname
  WHERE bug_id = @iv_bug_id.

IF sy-subrc = 0.
  COMMIT WORK.
  ev_status = '2'. " Assigned
  ev_dev_id = lv_best_dev.
  ev_message = 'Auto-assigned successfully'.
```

```
    " 6. Gửi Email thông báo
    CALL FUNCTION 'Z_BUG_SEND_EMAIL'
      EXPORTING
        iv_bug_id = iv_bug_id
        iv_event  = 'ASSIGN'.
  ENDIF.

ENDFUNCTION.
```

Nhấn **Activate**.

> ✅ **Checkpoint: SE37** → `Z_BUG_AUTO_ASSIGN`:

- Bug trong project có 1 dev → auto-assign, STATUS=2
- Bug trong project có 0 dev → STATUS='W' (Waiting)
- Dev đã bị soft-delete → không được assign

Bước B9: Tạo mới `Z_BUG_REASSIGN`

Mục tiêu: FM mới cho phép Manager chuyển bug sang dev khác.

> ⚠️ FM này **KHÔNG tồn tại** trong MVP — cần tạo hoàn toàn mới trong Function Group `ZBUG_FG`.

Vào **SE37** → nhập `Z_BUG_REASSIGN` → **Create** → Function Group `ZBUG_FG`.

Source code:

```
FUNCTION z_bug_reassign.
*"----------------------------------------------------------------------
*"*"Local Interface:
*"  IMPORTING
*"     VALUE(IV_BUG_ID) TYPE  ZDE_BUG_ID
*"     VALUE(IV_NEW_DEV_ID) TYPE  ZDE_USERNAME
*"     VALUE(IV_REASON) TYPE  ZDE_REASONS
*"     VALUE(IV_REQUESTED_BY) TYPE  ZDE_USERNAME
*"  EXPORTING
*"     VALUE(EV_STATUS) TYPE  ZDE_BUG_STATUS
*"     VALUE(EV_MESSAGE) TYPE  STRING
*"----------------------------------------------------------------------

  DATA: ls_bug      TYPE zbug_tracker,
        lv_role     TYPE char1,
        lv_count    TYPE i,
        lv_uname    TYPE sy-uname.
```

```
lv_uname = sy-uname.

" 1. Kiểm tra Bug tồn tại
SELECT SINGLE * FROM zbug_tracker INTO @ls_bug
  WHERE bug_id = @iv_bug_id AND is_del <> 'X'.
IF sy-subrc <> 0.
  ev_status = 'N'.
  ev_message = 'Bug not found'.
  RETURN.
ENDIF.

" 2. Kiểm tra quyền Manager
SELECT SINGLE role FROM zbug_users INTO @lv_role
  WHERE user_id = @lv_uname AND is_del <> 'X'.
IF lv_role <> 'M'.
  ev_status = 'N'.
  ev_message = 'Only Manager can reassign'.
  RETURN.
ENDIF.

" 3. Kiểm tra Dev mới có thuộc Project không (ZBUG_USER_PROJEC)
SELECT COUNT(*) FROM zbug_user_projec INTO @lv_count
  WHERE user_id = @iv_new_dev_id
    AND project_id = @ls_bug-project_id.
IF lv_count = 0.
  ev_status = 'N'.
  ev_message = 'New developer must belong to same project'.
  RETURN.
ENDIF.

" 4. Cập nhật Dev mới
UPDATE zbug_tracker SET dev_id = @iv_new_dev_id,
                        status = '2',
                        aenam  = @lv_uname
  WHERE bug_id = @iv_bug_id.

IF sy-subrc = 0.
  ev_status = '2'.
  COMMIT WORK.

  " Gửi Email thông báo REASSIGN
  CALL FUNCTION 'Z_BUG_SEND_EMAIL'
    EXPORTING
      iv_bug_id = iv_bug_id
      iv_event  = 'REASSIGN'.
ELSE.
  ev_status = 'N'.
  ROLLBACK WORK.
ENDIF.
```

```
ENDFUNCTION.
```

Nhấn **Activate**.

✅ **Checkpoint: SE37** → `Z_BUG_REASSIGN`:

- Manager reassign từ Dev1 sang Dev2 (cùng project) → Success, DEV_ID=Dev2
- Non-manager gọi → "Only Manager…"
- Dev2 không thuộc project → "New developer must belong to the same project"
- Bug ở status R → status reset về 2 (Assigned)

TỔNG KẾT PHASE B

Sau khi hoàn thành Phase B, bạn phải có:

- ☒ `Z_BUG_CHECK_PERMISSION` — hỗ trợ 12 actions (Bug + Project) + project membership
- ☒ `Z_BUG_CREATE` — validate Project + Severity + User membership + Config bug branch
- ☒ `Z_BUG_UPDATE_STATUS` — 9 status states + transition validation + evidence check + special fields (CLOSED_AT, VERIFY_TESTER_ID, APPROVED_BY)
- ☒ `Z_BUG_GOS_UPLOAD` / `Z_BUG_GOS_LIST` — upload/list file qua BDS
- ☒ `ZBUG_EMAIL_FORM` — SmartForm template HTML
- ☒ `Z_BUG_SEND_EMAIL` — CL_BCS + HTML + event-based recipients (CREATE, ASSIGN, REASSIGN, STATUS_CHANGE, REJECT)
- ☒ Soft delete logic áp dụng trong tất cả các FM
- ☒ `Z_BUG_AUTO_ASSIGN` — **updated**: IS_DEL filter + project membership + Waiting status fallback
- ☒ `Z_BUG_REASSIGN` — **MỚI**: Manager reassign dev + status reset + validation

Tổng FM count: 9 FMs (7 updated + 2 mới: GOS_LIST, REASSIGN)

👉 **Chuyển sang Phase C: Module Pool UI**